

Simulated Incident Response Investigation — [6/14/2026]

OBJECTIVE: Conduct a simulated security investigation on a Linux system, practicing log analysis, user account review, file system inspection, and incident documentation

SCENARIO: Investigation of repeated failed authentication attempts and privileged account activity

ENVIRONMENT: Ubuntu 26.04 VM

SECTION 1: SCENARIO SETUP

- Activity simulated:

- Multiple failed root authentication attempts
- Invalid SSH login attempts using a non-existent account
- Successful sudo activity
- Creation of a world-writable file in */tmp*
- Creation of a root-owned file containing test data

```
user@Ubuntu:~/scripts$ su - root
Password:
su: Authentication failure
user@Ubuntu:~/scripts$ su - root
Password:
su: Authentication failure
user@Ubuntu:~/scripts$ su - root
Password:
su: Authentication failure
user@Ubuntu:~/scripts$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
user@Ubuntu:~/scripts$ sudo ls /root
[sudo: authenticate] Password:
snap  vboxpostinstall.sh
user@Ubuntu:~/scripts$ sudo cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
```

```
sshd:x:973:65534:sshd user:/run/sshd:/usr/sbin/nologin
user@Ubuntu:~/scripts$ sudo whoami
root
user@Ubuntu:~/scripts$ sudo touch /tmp/suspicious_file.txt
user@Ubuntu:~/scripts$ sudo chmod 777 /tmp/suspicious_file.txt
user@Ubuntu:~/scripts$ echo "test data" | sudo tee /tmp/suspicious_file.txt
test data
```

- What I was looking for:

- Failed authentication attempts
- Invalid login attempts
- Privileged account activity
- Suspicious file creation
- Events that could later be identified through log analysis

- What I learned:

I learned that failed authentication attempts against the root account generate realistic authentication events that could be investigated through system logs. During the SSH testing phase, the specific terminal message was less important than the evidence written to auth.log because security investigations rely on recorded events rather than user-facing output. I also learned that a file containing data is more realistic than an empty file because investigators often review both file metadata and file contents when examining suspicious activity. The tee command was used to write data to the file while simultaneously displaying the output on the screen. This confirmed that the file was successfully created and populated. Finally, I learned that attackers often stage files in the /tmp directory because it is writable. Temporary files are also common there, and suspicious files can blend in with legitimate temp files. Many users and processes can also access the directory.

SECTION 2: SYSTEM OVERVIEW

- Active users observed:

- One active local session was identified through the w command.
- The session belonged to the user account on tty2

- Local user accounts found:

- root
- sync
- user
- testuser
- adminuser

```

user@Ubuntu:~/scripts$ uptime
 04:02:24 up 42 min,  1 user,  load average: 0.15, 0.04, 0.10
user@Ubuntu:~/scripts$ last reboot
reboot   system boot  7.0.0-22-generic Mon Jun 15 03:19 - still running
reboot   system boot  7.0.0-22-generic Sun Jun 14 03:38 - 10:13   (06:35)

wtmpdb begins Sat Jun 13 02:56:59 2026
user@Ubuntu:~/scripts$ who
user@Ubuntu:~/scripts$ w
 04:02:32 up 43 min,  1 user,  load average: 0.14, 0.04, 0.10
USER      TTY      FROM          LOGIN@  IDLE   JCPU   PCPU   WHAT
user      tty2      -             03:21   42:43   0.02s  0.02s  /usr/libexec/gnome-session-init-worker ubuntu
user@Ubuntu:~/scripts$ getent passwd | grep -v nologin | grep -v false
root:x:0:0:root:/root:/bin/bash
sync:x:4:65534:sync:/bin:/bin/sync
user:x:1000:1000:user:/home/user:/bin/bash
testuser:x:1001:1001:Test1,1,123,321,N/A:/home/testuser:/bin/bash
adminuser:x:1002:1002:Admin1,2,456,654,N/A:/home/adminuser:/bin/bash
user@Ubuntu:~/scripts$ getent group sudo
sudo:x:27:user

```

- Users with sudo access:

user

- System uptime and last reboot:

System uptime at the time of investigation was approximately 42 minutes

Current system boot occurred on Monday, June 15th, at 03:19 and was still running

Previous boot occurred on Sunday, June 14th, from 03:38 to 10:13

- What I learned:

I learned that active sessions and local accounts provide different information during an investigation. The *w* command identifies users currently interacting with the system.

getent passwd shows all accounts that exist. I also learned that *getent* is preferred over directly reading configuration files because it queries the system account databases and works with both local and directory-backed authentication sources. Reviewing sudo group membership also provided a quick method of identifying accounts with administrative privileges.

SECTION 3: LOGIN INVESTIGATION

- Failed login attempts found:

6 failed login attempts

- Invalid user events found:

3 invalid user events

- Timestamps observed:

```
user@ubuntu:~/scripts$ sudo tail -50 /var/log/auth.log
[sudo: authenticate] Password:
2026-06-15T03:39:39.156187+00:00 Ubuntu sshd-session[5867]: PAM 2 more authentication failures; logname=uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
2026-06-15T03:39:53.081332+00:00 Ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by user(uid=1000)
2026-06-15T03:39:53.081595+00:00 Ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-15T03:40:00.165668+00:00 Ubuntu unix_chkpwd[5954]: password check failed for user (root)
2026-06-15T03:40:00.165968+00:00 Ubuntu su: pam_unix(su-l:auth): authentication failure; logname=user uid=1000 euid=0 tty=/dev/pts/0 ruser=user rhost=
user=root
2026-06-15T03:40:14.935775+00:00 Ubuntu su[5950]: FAILED SU (to root) user on pts/0
2026-06-15T03:40:24.151882+00:00 Ubuntu unix_chkpwd[5963]: password check failed for user (root)
2026-06-15T03:40:24.152254+00:00 Ubuntu su: pam_unix(su-l:auth): authentication failure; logname=user uid=1000 euid=0 tty=/dev/pts/0 ruser=user rhost=
user=root
2026-06-15T03:40:26.478459+00:00 Ubuntu su[5958]: FAILED SU (to root) user on pts/0
2026-06-15T03:40:31.279601+00:00 Ubuntu unix_chkpwd[5970]: password check failed for user (root)
2026-06-15T03:40:31.280319+00:00 Ubuntu su: pam_unix(su-l:auth): authentication failure; logname=user uid=1000 euid=0 tty=/dev/pts/0 ruser=user rhost=
user=root
2026-06-15T03:40:33.752401+00:00 Ubuntu su[5966]: FAILED SU (to root) user on pts/0
2026-06-15T03:40:49.726242+00:00 Ubuntu sshd-session[5975]: Invalid user fakeuser from 127.0.0.1 port 40594
2026-06-15T03:40:52.545675+00:00 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): check pass; user unknown
2026-06-15T03:40:52.545793+00:00 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=12
7.0.0.1
```

```
user@ubuntu:~/scripts$ sudo grep "Failed password" /var/log/auth.log
2026-06-15T03:34:17.935076+00:00 Ubuntu sshd-session[5700]: Failed password for invalid user fakeuser from 127.0.0.1 port 59280 ssh2
2026-06-15T03:34:37.885432+00:00 Ubuntu sshd-session[5700]: Failed password for invalid user fakeuser from 127.0.0.1 port 59280 ssh2
2026-06-15T03:39:22.030877+00:00 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
2026-06-15T03:39:31.379477+00:00 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
2026-06-15T03:39:39.071727+00:00 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
2026-06-15T03:40:55.476690+00:00 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
2026-06-15T03:41:03.375339+00:00 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
2026-06-15T03:41:09.694996+00:00 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
2026-06-15T04:29:44.865997+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user/scripts ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/
log/auth.log
```

```
user@ubuntu:~/scripts$ sudo grep "Invalid user" /var/log/auth.log
2026-06-15T03:34:07.195647+00:00 Ubuntu sshd-session[5700]: Invalid user fakeuser from 127.0.0.1 port 59280
2026-06-15T03:39:17.400165+00:00 Ubuntu sshd-session[5867]: Invalid user fakeuser from 127.0.0.1 port 38740
2026-06-15T03:40:49.726242+00:00 Ubuntu sshd-session[5975]: Invalid user fakeuser from 127.0.0.1 port 40594
2026-06-15T04:29:59.167202+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user/scripts ; USER=root ; COMMAND=/usr/bin/grep Invalid user /var/log
/auth.log
```

- Pattern observed:

Clustered

- Commands used:

```
sudo tail -50 /var/log/auth.log
sudo grep "Failed password" /var/log/auth.log
sudo grep "Invalid user" /var/log/auth.log
sudo grep "authentication failure" /var/log/auth.log
sudo grep "Accepted password" /var/log/auth.log
sudo grep "sudo" /var/log/auth.log
```

- What I learned:

I learned that reviewing the raw log with *tail* before filtering results with *grep* can provide important context during an investigation. The raw log showed surrounding authentication activity and helped establish the sequence of events. *grep* isolated specific evidence such as failed password attempts, invalid user events, and authentication failures. I also learned that direct analysis of *auth.log* provides additional information like detailed timestamps, source addresses, and authentication outcomes. The authentication attempts observed were clustered together, which can indicate automated activity or rapid manual login attempts.

SECTION 4: FILE SYSTEM INVESTIGATION

- ls -l output for suspicious file:

```
-rwxrwxrwx 1 root root 10 Jun 15 03:42 /tmp/suspicious_file.txt
```

```

user@Ubuntu:~/scripts$ ls -l /tmp/suspicious_file.txt
-rwxrwxrwx 1 root root 10 Jun 15 03:42 /tmp/suspicious_file.txt
user@Ubuntu:~/scripts$ ls -lt /tmp/
total 4
drwx----- 3 root root 60 Jun 15 04:14 systemd-private-bc7080d225e144d090ea0ae0287030fb-fwupd.service-hFmEEU
-rwxrwxrwx 1 root root 10 Jun 15 03:42 suspicious_file.txt
drwx----- 5 root root 100 Jun 15 03:23 snap-private-tmp
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-upower.service-jo0y1z
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-colord.service-dPBGMV
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-power-profiles-daemon.service-rvdTjk
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-ModemManager.service-aDgXxI
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-systemd-logind.service-WEdHQX
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-switcheroo-control.service-O5SNxx
drwx----- 3 root root 60 Jun 15 03:19 systemd-private-bc7080d225e144d090ea0ae0287030fb-polkit.service-omjNtJ
drwx----- 3 root root 60 Jun 15 03:19 systemd-private-bc7080d225e144d090ea0ae0287030fb-chrond.service-B2ek1V
user@Ubuntu:~/scripts$ find /tmp -perm -o+w -type f
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-fwupd.service-hFmEEU': Permission denied
/tmp/suspicious_file.txt
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-upower.service-jo0y1z': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-colord.service-dPBGMV': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-power-profiles-daemon.service-rvdTjk': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-ModemManager.service-aDgXxI': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-systemd-logind.service-WEdHQX': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-switcheroo-control.service-O5SNxx': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-polkit.service-omjNtJ': Permission denied
find: '/tmp/systemd-private-bc7080d225e144d090ea0ae0287030fb-chrond.service-B2ek1V': Permission denied
find: '/tmp/snap-private-tmp': Permission denied

```

- Owner and group of suspicious file:

Owner: root

Group: root

- File contents observed:

“test data”

- Suspicious files found via find:

/tmp/suspicious_file.txt

- Permissions observed:

777 (world writable)

- Commands used:

ls -l /tmp/suspicious_file.txt

ls -lt /tmp/

find /tmp -perm -o+w -type f

ps aux

ps -eo user,pid,cmd

sudo ss -tlnp

sudo grep “COMMAND” /var/log/auth.log

```

user@Ubuntu:~/scripts$ sudo grep "COMMAND" /var/log/auth.log
2026-06-13T02:57:34.650739+00:00 Ubuntu pexec[5319]: user: Executing command [USER=root] [TTY=unknown] [CWD=/home/user] [COMMAND=/usr/lib/update-noti
fier/package-system-locked]
2026-06-13T03:08:25.443866+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/apt update
2026-06-13T03:08:35.908857+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/apt upgrade -y
2026-06-13T03:09:17.390165+00:00 Ubuntu pexec[7879]: user: Executing command [USER=root] [TTY=unknown] [CWD=/home/user] [COMMAND=/usr/lib/update-noti
fier/package-system-locked]
2026-06-13T03:12:07.939471+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/sbin/adduser testuser
2026-06-13T03:14:21.386554+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/sbin/adduser adminuser
2026-06-13T03:16:44.300130+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/sbin/groupadd engineers
2026-06-13T03:16:54.503304+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/sbin/usermod -aG engineers testuser
2026-06-13T03:17:02.007050+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/sbin/usermod -aG engineers adminuser
2026-06-13T03:35:49.676567+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/chown testuser:engineers /home/use
r/public_folder/testfile.txt
2026-06-13T03:48:36.605206+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/chown testuser:engineers /home/use
r/public_folder/testfile.txt
2026-06-13T03:51:08.672446+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/chown testuser:engineers /home/use
r/public_folder/testfile.txt
2026-06-13T03:58:06.267450+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/apt install openssh-server -y
2026-06-13T03:59:41.270686+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-06-13T04:00:01.308536+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/systemctl list-unit-files
2026-06-13T04:00:13.144603+00:00 Ubuntu sudo: user : TTY=/dev/pts/0 ; PWD=/home/user ; USER=root ; COMMAND=/usr/bin/systemctl stop ssh

```

- What I learned:

I learned that the `-rwxrwxrwx` permission string corresponds to 777 permissions. Because the suspicious file was owned by root while remaining world writable, it showed a security risk since any user could modify a privileged file. I also learned that when reviewing processes, `ps -eo user,pid,cmd` provided a cleaner investigative view than manually filtering process output. This is because the `-e` flag selects all processes and the `-o` flag specifies exactly which fields to display. Finally, I learned that directly using `grep` on a file is cleaner than piping output from `cat` into `grep` because it adds an unnecessary command without providing additional functionality. The `find` command returned Permission denied for `systemd-private` directories. The suspicious file was still correctly identified despite these permission errors.

SECTION 5: NETWORK INVESTIGATION

- Commands used:

```
sudo ss -tulnp
```

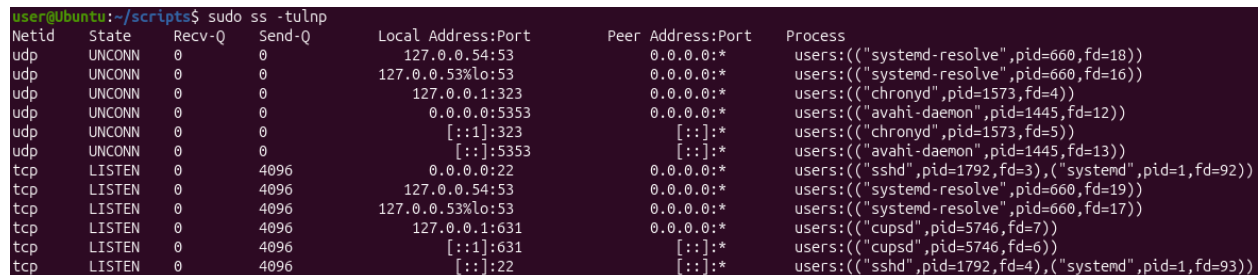
```
ip a
```

```
ip r
```

```
nslookup google.com
```

```
sudo journalctl -u ssh --since "3 hours ago"
```

```
sudo ss -tnp | grep ESTABLISHED
```



Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	users(("systemd-resolve",pid=660,fd=18))
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	users(("systemd-resolve",pid=660,fd=16))
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*	users(("chronyd",pid=1573,fd=4))
udp	UNCONN	0	0	0.0.0.0:5353	0.0.0.0:*	users(("avahi-daemon",pid=1445,fd=12))
udp	UNCONN	0	0	:::1:323	::::*	users(("chronyd",pid=1573,fd=5))
udp	UNCONN	0	0	:::5353	::::*	users(("avahi-daemon",pid=1445,fd=13))
tcp	LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*	users(("sshd",pid=1792,fd=3),("systemd",pid=1,fd=92))
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	users(("systemd-resolve",pid=660,fd=19))
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	users(("systemd-resolve",pid=660,fd=17))
tcp	LISTEN	0	4096	127.0.0.1:631	0.0.0.0:*	users(("cupsd",pid=5746,fd=7))
tcp	LISTEN	0	4096	:::1:631	::::*	users(("cupsd",pid=5746,fd=6))
tcp	LISTEN	0	4096	:::22	::::*	users(("sshd",pid=1792,fd=4),("systemd",pid=1,fd=93))

- Open ports found:

Port 22 (SSH)

Port 53 (systemd-resolved DNS)

Port 323 (Chrony NTP)

Port 631 (CUPS printing service)

- Active connections:

No established TCP connections were observed.

- SSH journal findings:

SSH logs showed repeated invalid user events, failed password attempts, authentication failures, and connection closures involving the username "fakeuser" originating from localhost (127.0.0.1).


```

user@Ubuntu:~/scripts$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:47:d6:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 77800sec preferred_lft 77800sec
    inet6 fd17:625c:f037:2:a00:27ff:fe47:d601/64 scope global dynamic noprefixroute
        valid_lft 85913sec preferred_lft 13913sec
    inet6 fe80::a00:27ff:fe47:d601/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
user@Ubuntu:~/scripts$ ip r
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
user@Ubuntu:~/scripts$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.40.238
Name:   google.com
Address: 2607:f8b0:4006:80f::200e

```

```

user@Ubuntu:~/scripts$ sudo journalctl -u ssh --since "3 hours ago"
Jun 15 03:20:22 Ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 15 03:20:23 Ubuntu sshd[1792]: Server listening on 0.0.0.0 port 22.
Jun 15 03:20:23 Ubuntu sshd[1792]: Server listening on :: port 22.
Jun 15 03:20:23 Ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jun 15 03:34:07 Ubuntu sshd-session[5700]: Invalid user fakeuser from 127.0.0.1 port 59280
Jun 15 03:34:14 Ubuntu sshd-session[5700]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:34:14 Ubuntu sshd-session[5700]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Jun 15 03:34:17 Ubuntu sshd-session[5700]: Failed password for invalid user fakeuser from 127.0.0.1 port 59280 ssh2
Jun 15 03:34:35 Ubuntu sshd-session[5700]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:34:37 Ubuntu sshd-session[5700]: Failed password for invalid user fakeuser from 127.0.0.1 port 59280 ssh2
Jun 15 03:34:43 Ubuntu sshd-session[5700]: Connection closed by invalid user fakeuser 127.0.0.1 port 59280 [preauth]
Jun 15 03:34:43 Ubuntu sshd-session[5700]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Jun 15 03:39:17 Ubuntu sshd-session[5867]: Invalid user fakeuser from 127.0.0.1 port 38740
Jun 15 03:39:19 Ubuntu sshd-session[5867]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:39:19 Ubuntu sshd-session[5867]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Jun 15 03:39:22 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
Jun 15 03:39:28 Ubuntu sshd-session[5867]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:39:31 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
Jun 15 03:39:36 Ubuntu sshd-session[5867]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:39:39 Ubuntu sshd-session[5867]: Failed password for invalid user fakeuser from 127.0.0.1 port 38740 ssh2
Jun 15 03:39:39 Ubuntu sshd-session[5867]: Connection closed by invalid user fakeuser 127.0.0.1 port 38740 [preauth]
Jun 15 03:39:39 Ubuntu sshd-session[5867]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Jun 15 03:40:49 Ubuntu sshd-session[5975]: Invalid user fakeuser from 127.0.0.1 port 40594
Jun 15 03:40:52 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:40:52 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Jun 15 03:40:55 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
Jun 15 03:41:00 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:41:03 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
Jun 15 03:41:07 Ubuntu sshd-session[5975]: pam_unix(sshd:auth): check pass; user unknown
Jun 15 03:41:09 Ubuntu sshd-session[5975]: Failed password for invalid user fakeuser from 127.0.0.1 port 40594 ssh2
Jun 15 03:41:09 Ubuntu sshd-session[5975]: Connection closed by invalid user fakeuser 127.0.0.1 port 40594 [preauth]
Jun 15 03:41:09 Ubuntu sshd-session[5975]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1

```

- What I learned:

I learned that network investigation involves reviewing listening services, network configuration, DNS functionality, and active connections. The `ss -tulnp` command identified services listening for network traffic. The `ip a` and `ip r` commands provided information about the system's network configuration and routing. DNS functionality was verified using `nslookup`. Although no established TCP connections were present during

the investigation, their absence indicated there were no ongoing network sessions at the time of review.

SECTION 6: INCIDENT REPORT

- Report saved at:

~/incident_reports/incident_report_2026-06-15.txt

```
user@Ubuntu:~/scripts$ ls -la ~/incident_reports/
total 16
drwxrwxr-x  2 user user 4096 Jun 15 07:00 .
drwxr-x--- 20 user user 4096 Jun 15 06:03 ..
-rw-rw-r--  1 user user 5918 Jun 15 07:00 incident_report_2026-06-15.txt
```

- Key findings documented:

Failed authentication attempts, invalid user events, privileged account activity, suspicious file creation, file permission review, and network activity findings were documented in the incident report.

- Timestamps documented in findings:

Authentication, SSH, and privileged activity timestamps were documented throughout the findings section and in the incident report.

- Source of attempts documented as localhost:

Log analysis identified localhost (127.0.0.1) as the source of the simulated authentication attempts.

- Recommendations made:

Monitor failed login attempts, review sudo access, restrict world-writable file permissions, implement automated log monitoring, and perform regular user account and session audits.

```
user@Ubuntu:~/scripts$ sudo rm /tmp/suspicious_file.txt
[sudo: authenticate] Password:
user@Ubuntu:~/scripts$ ls -lt /tmp/
total 0
drwx----- 6 root root 120 Jun 15 06:01 snap-private-tmp
drwx----- 3 root root 60 Jun 15 04:14 systemd-private-bc7080d225e144d090ea0ae0287030fb-fwupd.service-hFnEEU
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-upower.service-jo0y1z
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-colord.service-dPBGMV
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-power-profiles-daemon.service-rvdTjk
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-ModemManager.service-aDgXxI
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-systemd-logind.service-WEdHQX
drwx----- 3 root root 60 Jun 15 03:20 systemd-private-bc7080d225e144d090ea0ae0287030fb-switcheroo-control.service-05SNxx
drwx----- 3 root root 60 Jun 15 03:19 systemd-private-bc7080d225e144d090ea0ae0287030fb-polkit.service-onjNtJ
drwx----- 3 root root 60 Jun 15 03:19 systemd-private-bc7080d225e144d090ea0ae0287030fb-chrony.service-B2ek1V
```

- What I learned about documentation:

I learned that effective incident documentation requires organizing evidence into a clear investigation timeline, documenting findings with supporting log evidence, identifying the source and scope of activity, and providing actionable recommendations.

CHALLENGES:

The command `sudo ss -tnp | grep ESTABLISHED` did not return any results. This initially appeared unexpected, but further analysis revealed that no active TCP sessions were present at the time of investigation.

SKILLS DEMONSTRATED:

- Log analysis
- Incident response methodology
- Authentication event investigation
- File system forensics
- Network activity review
- Timeline reconstruction
- Technical report writing
- Root cause analysis
- Evidence documentation